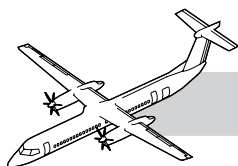


CHAPTER 51

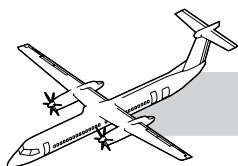
STANDARD PRACTICES AND STRUCTURES

CONTENTS

	Page
51-00-00 INTRODUCTION	51-1
General	51-1
Overview.....	51-3
Training Information Point.....	51-5
Critical Zones.....	51-9
Safety Precautions.....	51-10
Environment Protection.....	51-10
51-22-00 BONDING WITH ADHESIVE	51-11
General	51-11
Procedures.....	51-11
Typical Sealing Methods	51-13
Alternate Fast Cure Sealant.....	51-19
Dispatch of the Aircraft with Uncured or Missing Sealant.....	51-19
51-25-00 APPLICATION OF AIRCRAFT PAINT	51-23
Introduction	51-23
General	51-23
Removal of Damaged Runway Protective Tape.....	51-23
Special Tooling	51-23
Safety Precautions.....	51-24
51-80-00 ELECTRICAL BONDING.....	51-27

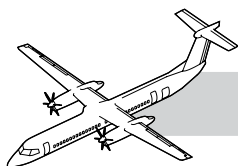


	Page
Introduction	51-27
Bonding Check - Resistance	51-27
Resistance Measurement	51-29
Application of HumiSeal® 1B15	51-30
Bonding Check of the Ice Detector Probe (IDP)	51-31
General	51-31
Damage Reporting	51-32
51-00-00 SPECIAL TOOL & TEST EQUIPMENT	51-33
51-00-00 MAINTENANCE PRACTICES	51-33



ILLUSTRATIONS

Figure	Title	Page
51-1	Aircraft Secondary Structure	51-2
51-2	Aircraft Composite Structures	51-4
51-3	Critical and Non-Critical Zones	51-8
51-4	Typical Sealing Methods.....	51-12
51-5	Sealing for Aerodynamic Smoothness.....	51-14
51-6	How to Apply Sealant with a Brush	51-15
51-7	Wet Installation of Permanent Fasteners.....	51-16
51-8	Dome Sealing Anchor Nut	51-17
51-9	Accelerated Cure Cycles for Original Sealing Compounds	51-18
51-10	Unpaved Runway Protective Tape Application	51-22
51-11	Bonding Strap.....	51-26
51-12	Resistance Measurement - De-Icing Boot/Airframe	51-28



CHAPTER 51 STANDARD PRACTICES AND STRUCTURES



51-00-00 INTRODUCTION

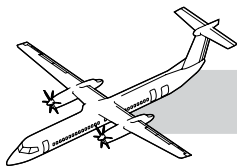
Most of the Dash 8 Series 400 aircraft structure is made from aluminium alloys. Alloy steels, corrosion-resistant steels, titanium, organic-fibre laminates and composites are used where necessary for strength, long serviceable life or heat protection. To decrease weight, honeycomb structures and chemically-milled panels are used in many locations.

GENERAL

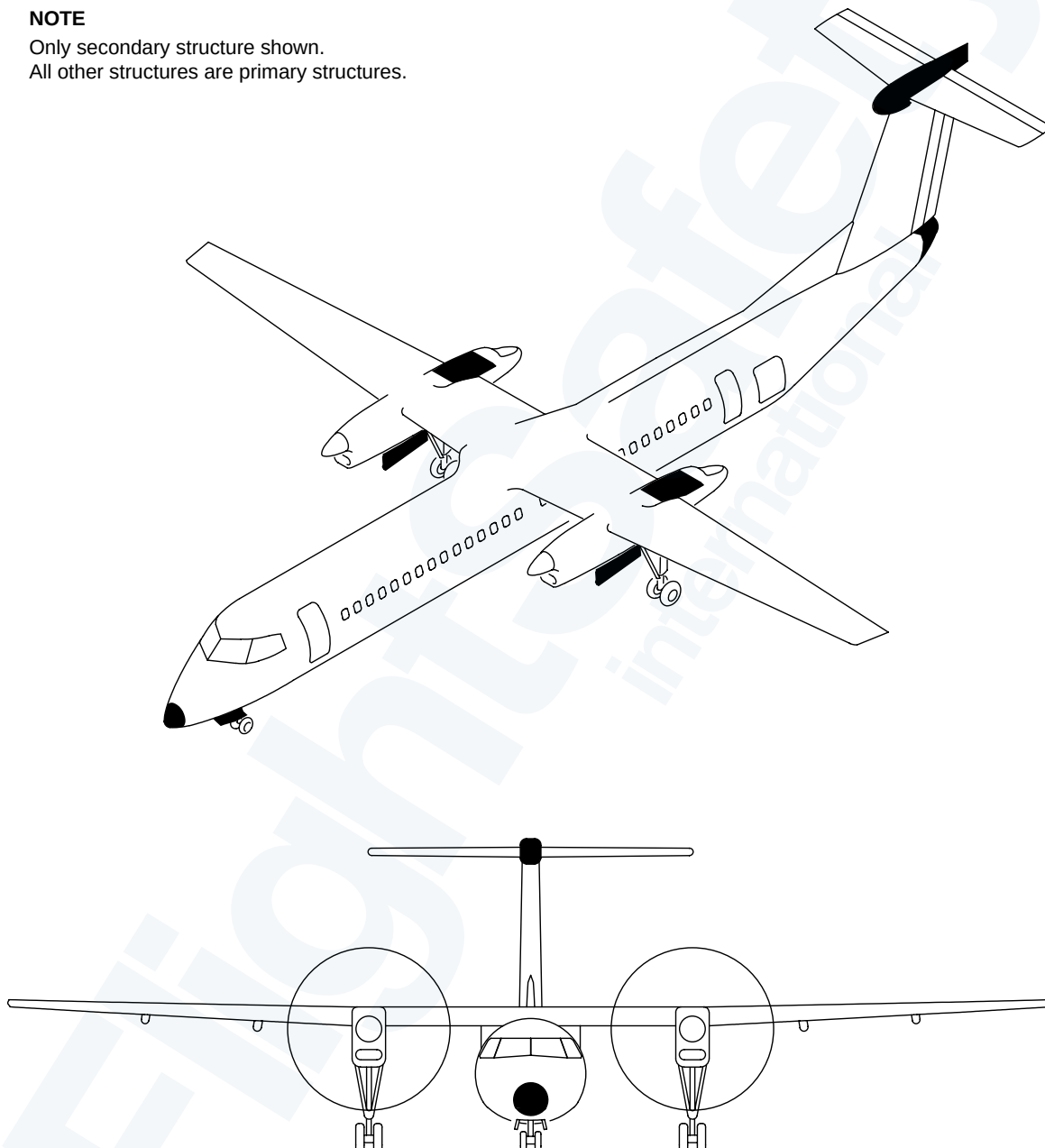
The airframe structure is made to be damage-tolerant against damage that results from fatigue, corrosion or accidental damage during the life of the aircraft. After damage has occurred (which the flight or maintenance crews cannot easily see), the remaining serviceable structure will be sufficiently strong to transmit usual operational loads until the damage is found and repaired without:

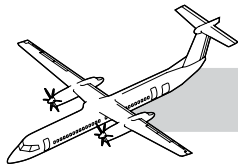
- Dangerous failure
- Too much structural deformation.

This is done by the usage of damage-resistant materials and multiple load-path construction together with scheduled structural inspection. The inspection frequency is found by damage tolerance analysis. For a description of the concept of damage tolerance, refer to Aircraft Damage Tolerance Certification, SRM TOPIC 51-05-00-00-01.

**LEGEND** Secondary structure**NOTE**

Only secondary structure shown.
All other structures are primary structures.

**Figure 51-1. Aircraft Secondary Structure**



OVERVIEW

The aircraft is a metal high winged monoplane with fully cantilever wings and horizontal stabilizer surfaces, a semi-monocoque fuselage and a fully retractable tricycle landing gear. Power is provided by two turbo-prop engines in wing mounted nacelles. A large portion of the skin panels are bonded assemblies consisting of a skin, stringers and doublers, or skin sandwich with a honeycomb core.

Refer to Figure 51-1. Aircraft Secondary Structure.

The aircraft structure is divided into primary and secondary classes to show structural importance. The classification of the structure will help you make decisions about how and when to repair damage. The primary and secondary structural identity of fittings and attachments is the same as the structures to which they are attached.

Primary Structure: The function of primary structure is to transmit loads that are caused by:

- Pressurization
- Wind gusts
- Flight manoeuvres
- Take-off and landings
- Ground manoeuvres.

The primary structure is divided into structural significant items (SSIs). The structural significant items are those structural parts that are the most important in the transmission of flight, ground and pressurization loads. Failure in the structural significant items of the primary structure can possibly cause structural collapse, loss of control or loss of motive power.

NOTE

When repairs are made to structural significant items, you must make sure that the repair is damage-tolerant.

For the necessary data as related to airworthiness for repairs that are done to the structural significant item, refer to Structural Repairs 51-15-00.

It is possible that all or part of a SSI is classified as an Airworthiness Limitation Item (ALI). The ALIs are those parts of the structure where mandatory maintenance and inspection are necessary to prevent the loss of airworthiness due to fatigue cracking. Some ALIs are replaced before fatigue cracks can occur. The remaining ALIs are inspected for cracks with a time controlled inspection program. The ALIs are identified in the Maintenance Program Document (MPD).

Secondary Structure: The primary structure can contain auxiliary parts that are secondary structure. Secondary structures are structures made to:

- Help the primary structure transmit aerodynamic loads
- Keep the aerodynamic shape
- Attach equipment.

Loads applied to the secondary structure are transmitted to the primary structure. Damage to the secondary structure can possibly cause unwanted effects to the primary structure, engine performance, aerodynamic qualities and/or surface stiffness. A careful analysis must be made of damage to secondary structure for these unwanted effects.

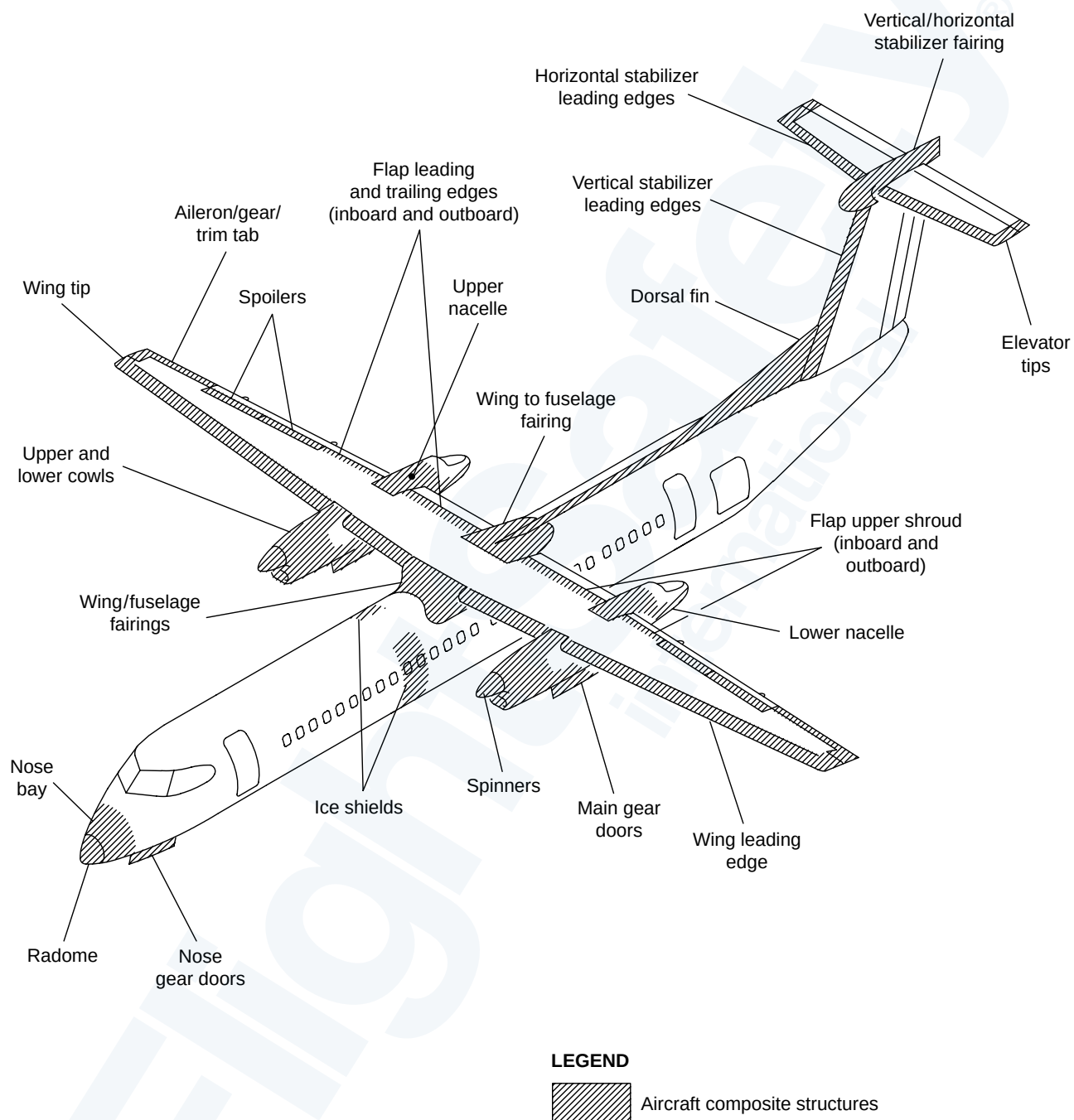
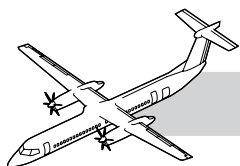
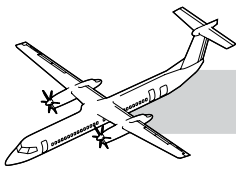


Figure 51-2. Aircraft Composite Structures



Refer to Figure 51-2. Aircraft Composite Structures.

A large portion of skin panels, leading edges, shrouds, fairings, ice shields and other structures are made of composite materials.

Do not use standard practices for any composite repairs to the aircraft. Refer to the applicable Chapter, Section, Subject in Structural Repair Manual, PSM 1-84-3 for specific repairs to composites or contact Bombardier Aerospace Structures Engineering.

TRAINING INFORMATION POINT

Damage is a change in cross-sectional area, separation (fully or partially), or permanent distortion of a structural part.

Damage to the aircraft can be caused by collision, fatigue, fire or corrosion.

Damage can also be caused by operation of the aircraft at a higher stress level than permitted by engineering limits.

Damage is divided into three distinct classifications:

- Allowable damage
- Repairable damage
- Damage repaired by replacement of affected parts.

Allowable damage: This is defined as light damage in the form of dents, nicks, scratches or surface corrosion, the extent of which has no measurable effect on the required strength and durability of the component, its aerodynamic qualities and intended function or performance. Such damage must be free from cracks. This type of damage is divided into two groups:

Corrective procedures or repairs are not necessary. Only small corrective procedures or repairs are necessary, such as:

- Removal of nicks and scratches
- Removal of light surface corrosion
- Corrosion preventive treatment.

Repairable damage: This is defined as damage exceeding the limits of allowable damage and may be repaired by one of the types of repair defined in each chapter, generic repair manual or by a specific approved repair scheme supplied by Bombardier Aerospace Regional Aircraft.

Repair by part replacement: Replacement of parts is in general the most satisfactory repair procedure. When the repair is correctly done, the area has the same strength, serviceable life, weight and configuration as it had before it was damaged. This can also be more economical than the repair of large areas of damage. Replacement of parts is necessary when:

- Parts are damaged and not economical to repair
- Damaged parts are too small for alternate means of repair
- Forged, cast and machined fittings are damaged more than the allowable damage limits
- After a repair to highly-stressed parts, the allowance for safety will not be sufficient
- Parts that have their strength decreased by the effect of fire or heat, metal becomes soft or weakened.

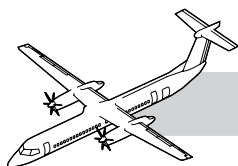
The following are definitions to the type of damage, given as an aid to damage identification and classification.

Dent:

A dent is normally a damage area which is depressed or deformed with respect to its normal contour. The area boundaries are smooth and uncracked. There is no change in the cross-sectional area of the material.

Abrasion:

An abrasion is a damage area of any size which results in a sectional area change in the material due to scuffing, rubbing, scrapping or other surface erosion.

**Gouge:**

A gouge is a continuous sharp or smooth groove in the surface of the material. A loss of material or cross-sectional area is associated with a gouge.

Nick:

A nick is a localized gouge with sharp edges. A series of nicks in a line pattern are considered to be equivalent to a gouge.

Scratch:

A scratch is a line of damage, usually caused by contact with a very sharp object and does not penetrate the full thickness of the material. A scratch which penetrates the full thickness is considered a crack.

Crack:

A crack is a partial or complete fracture in the material and is usually in the form of an irregular line.

Puncture:

A puncture is a localized hole or perforation which penetrates the full thickness of the material.

Corrosion:

Corrosion is a damage area due to exposure of unprotected metal surfaces, environmental conditions, or due to electrochemical action. Depth of such damage must be determined after removal of the corrosion.

Void:

Local absence of bonding between any two bonded elements or local absence of core in a sandwich structure.

Wrinkle:

Local waviness of a surface.

Buckle:

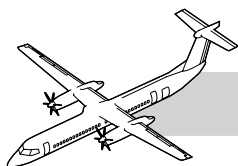
A surface that is warped or distorted.

Crease:

A depression in the surface of material caused by contact with another object.

Delamination:

Separation of layers in a bonded assembly.



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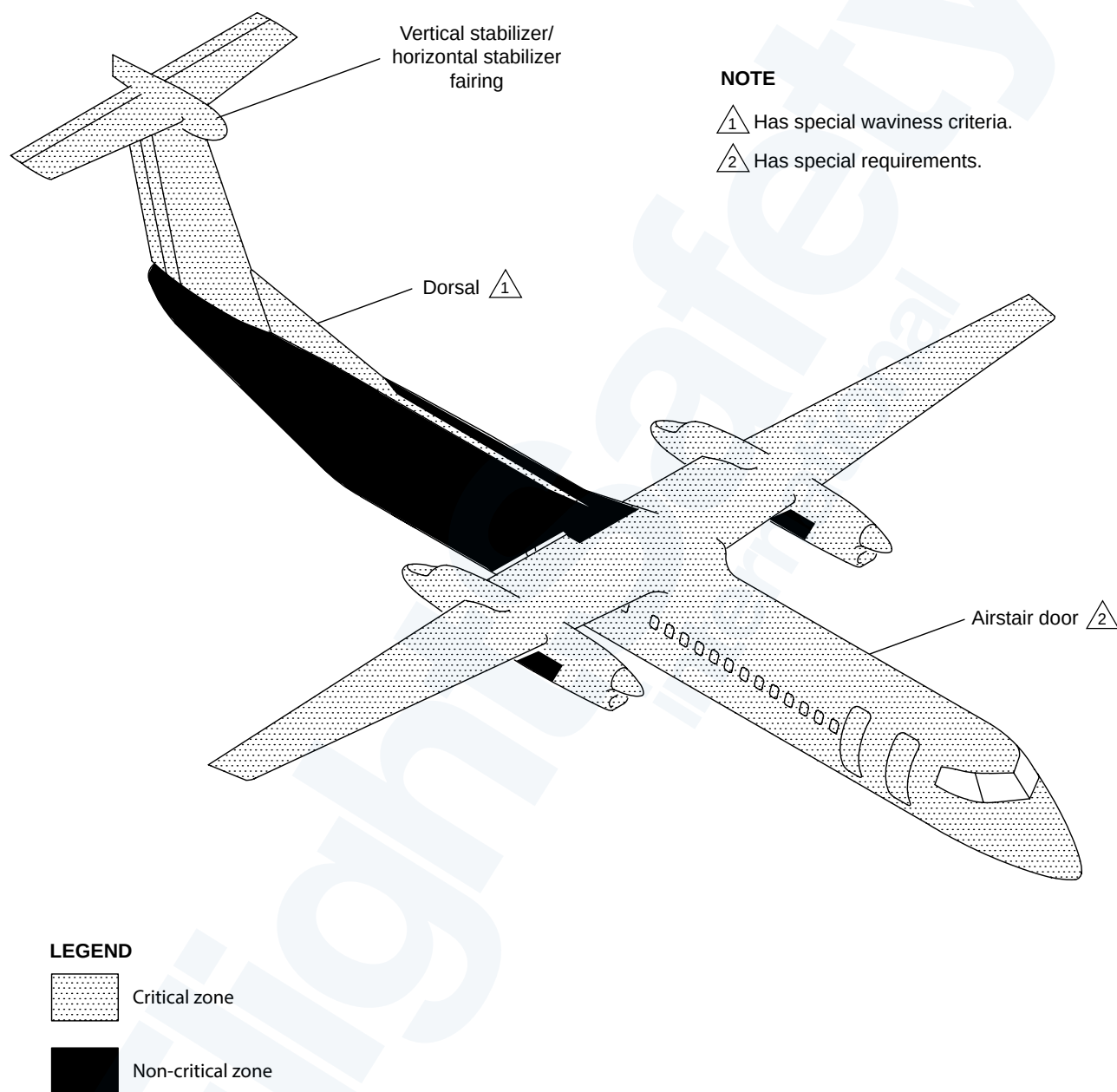
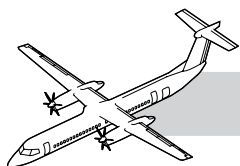
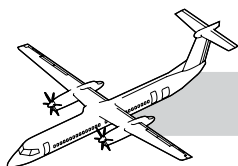


Figure 51-3. Critical and Non-Critical Zones

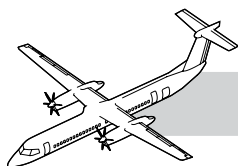


CRITICAL ZONES

NOTES

Refer to Figure 51-3. Critical and Non-Critical Zones.

This section specifies the critical zones in the Dash 8, Series 400 aircraft. Any deviation or repairs must be approved by Bombardier Aerospace.



SAFETY PRECAUTIONS

Use approved protection equipment such as rubber gloves, safety glasses, rubber apron and respirators.

Avoid skin contact with solvents or sealants.

Should accidental eye contact with solvents/adhesives occur, flush immediately with large quantities of water and seek immediate medical attention.

Keep your work area clean and have a good flow of clean air.

Provide sufficient ventilation when working in confined spaces, avoid prolonged breathing of solvent and sealant fumes.

Keep CPC, solvents and sealants away from fire and other sources of ignition.

No sparks, open flames or heat lamps in the work area.

Ground the aircraft, all the major components, and the equipment.

Do not eat or drink, or smoke in areas where solvents are being handled.

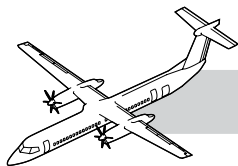
If ingestion of solvent occurs, get medical aid immediately.

If irritation occurs, get medical aid. Solvents are poisonous.

Refer to the applicable manufacturer's Material Safety Data Sheet (MSDS) for specific data on any of the materials used.

ENVIRONMENT PROTECTION

Many of the materials used can cause damage to the environment. When you dispose of these materials, you must follow the local government health and safety regulations.



51-22-00 BONDING WITH ADHESIVE

NOTES

GENERAL

Bonding or attaching different parts or materials to each other with adhesives.

Procedures

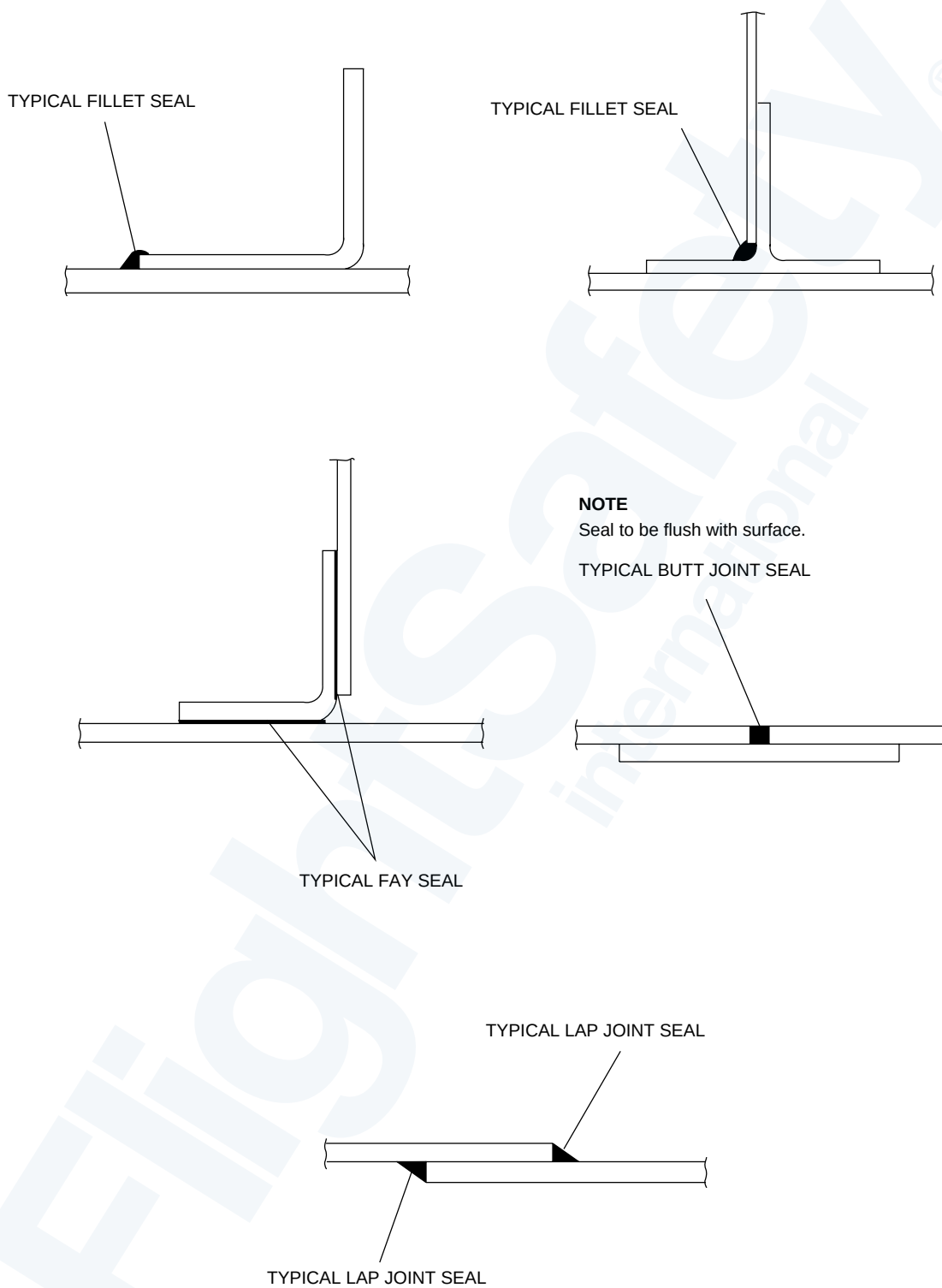
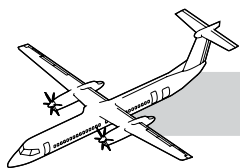
- Read and obey all instructions carefully
- Do not touch clean parts with bare hands. Use lint free cotton gloves
- Make sure that the shelf life of the adhesive materials have not expired
- Read and obey all manufacturer's instructions.

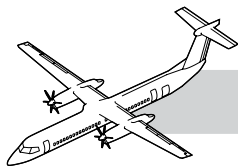
NOTE

Porous surfaces may require two or more coats to ensure that sufficient adhesive remains on the surface. Allow each preceding coat to air dry before the next coat is applied.

NOTE

After adhesive has dried to a tack-free condition, parts which are not to be bonded immediately should be covered with clean Kraft paper.

**Figure 51-4. Typical Sealing Methods**



TYPICAL SEALING METHODS

CAUTION

Refer to:

- Figure 51-4. Typical Sealing Methods.
- Figure 51-5. Sealing for Aerodynamic Smoothness.
- Figure 51-6. How to Apply Sealant with a Brush.
- Figure 51-7. Wet Installation of Permanent Fasteners.
- Figure 51-8. Dome Sealing Anchor Nut.

Typical sealing methods used with repairs to the aircraft structure are illustrated in figures 51-4 through 51-8.

Acceptable sealant use and locations to be used are listed in the SRM.

Follow the instructions given by the manufacturer of the sealant as well as the instructions outlined in the AMM.

Types of Sealants:

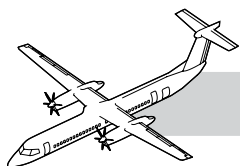
- Fuel Tank Sealants
- Firewall Sealants
- Fuel Tank Corrosion Preventive Primer Sealant
- Aerodynamic and Pressure Sealing Sealants
- Faying Surface Sealants For General Structure
- Alternate Fast Cure Sealants.

Refer to the Tables in the SRM for different sealants, manufacturers, specifications and applications.

NOTE

During sealing operations, make sure the drain holes in the structure remain unobstructed and free from sealant.

The sealant around the fuel tank dome anchor nuts are classified as Critical Design Configuration Control Limitation (CDCCL) items with the purpose of providing protection in the event of a lightning strike. The integrity of the sealant around the dome anchor nuts must be maintained to make sure that unsafe conditions do not develop by maintenance, modification or repairs. For details, refer to the SRM.



DASH 8 Q400 MAINTENANCE TRAINING MANUAL

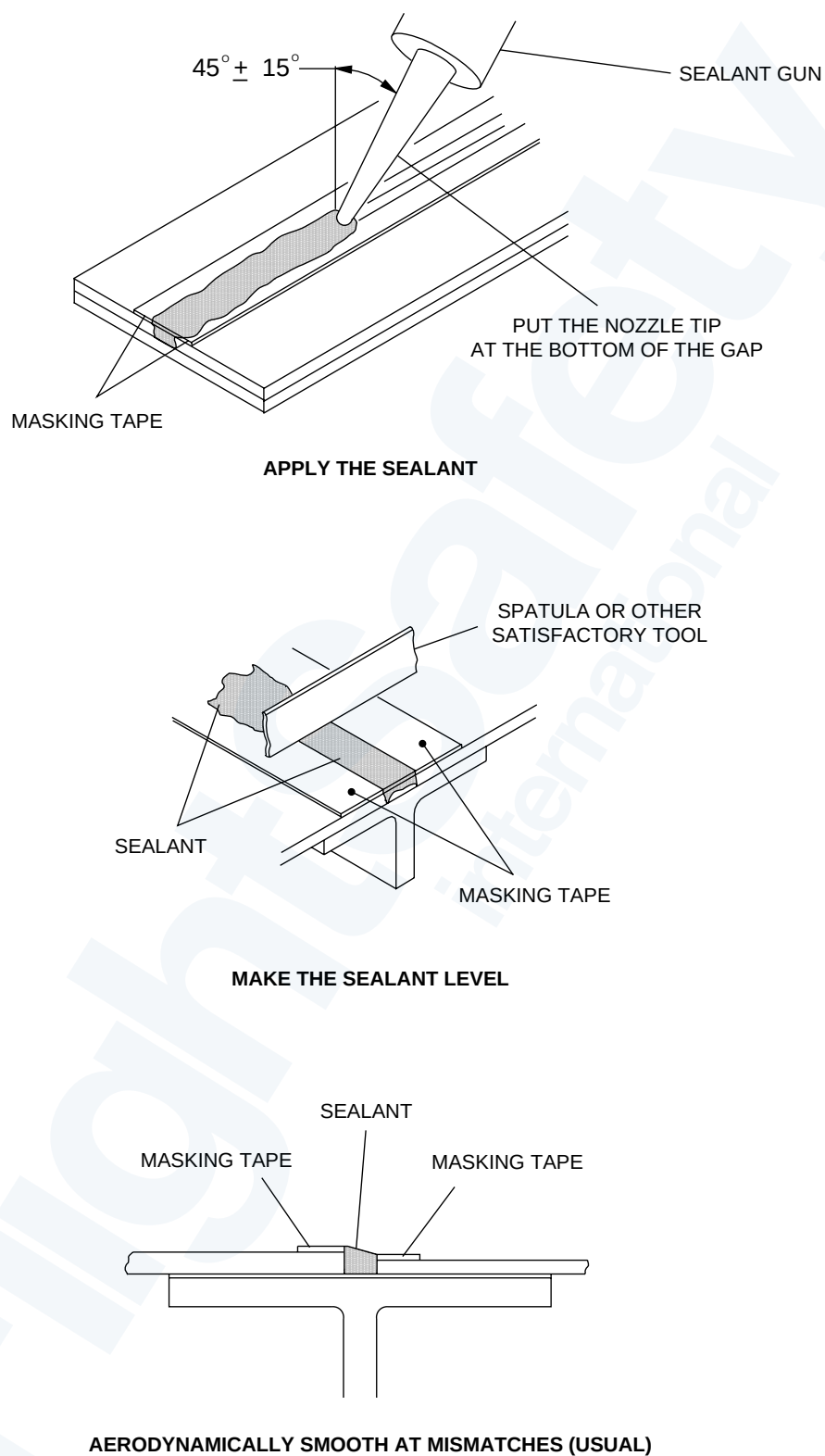


Figure 51-5. Sealing for Aerodynamic Smoothness

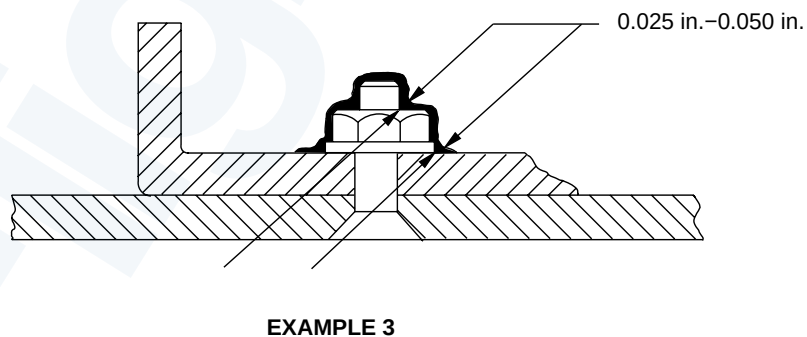
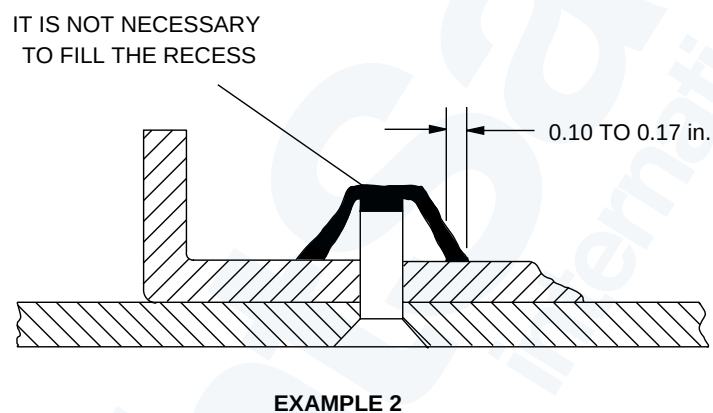
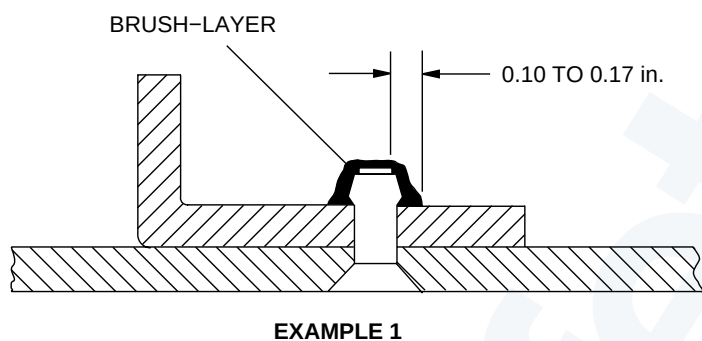
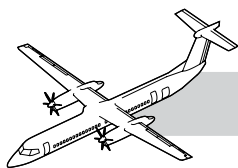
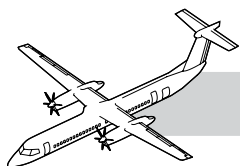


Figure 51-6. How to Apply Sealant with a Brush



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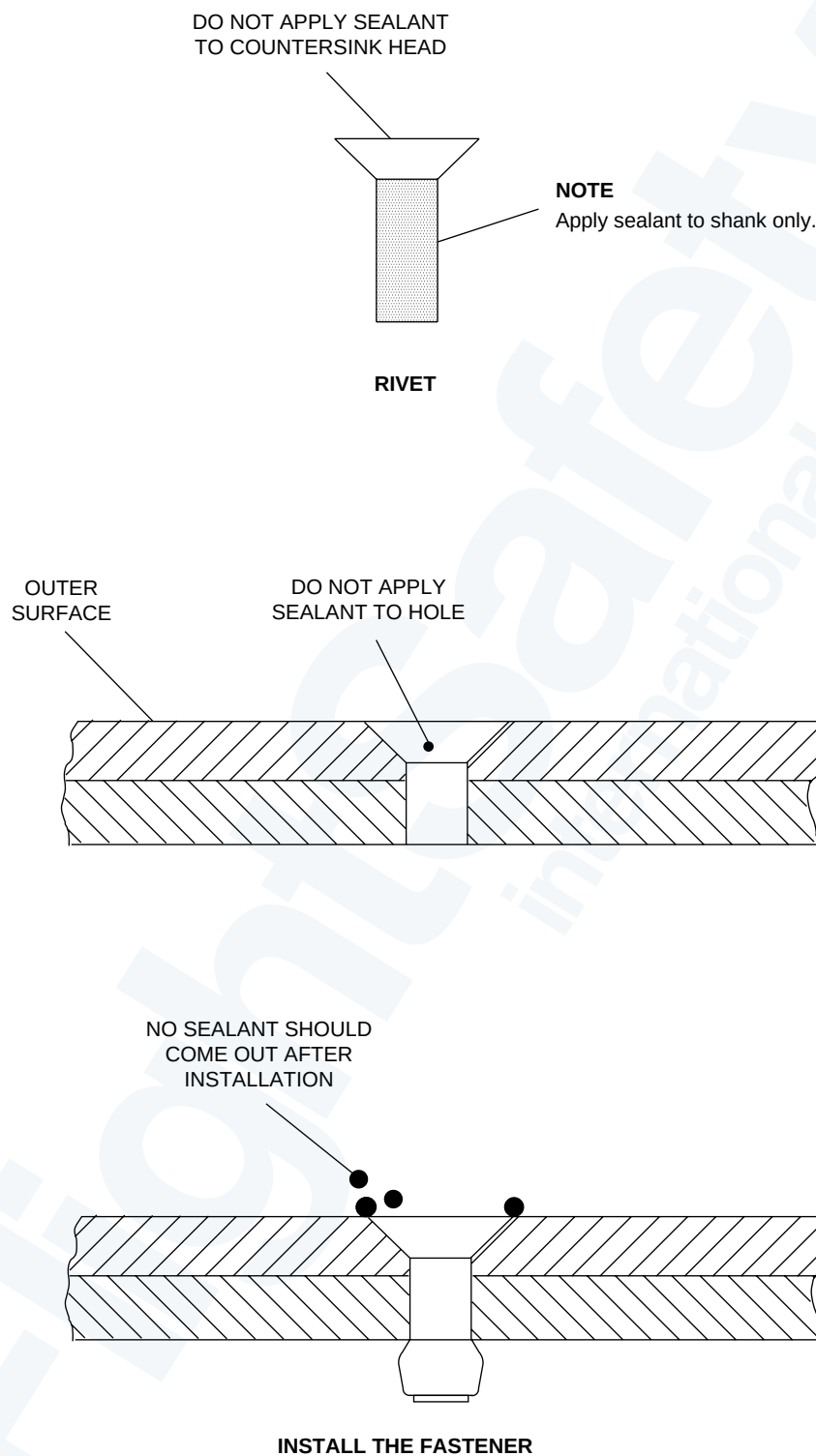
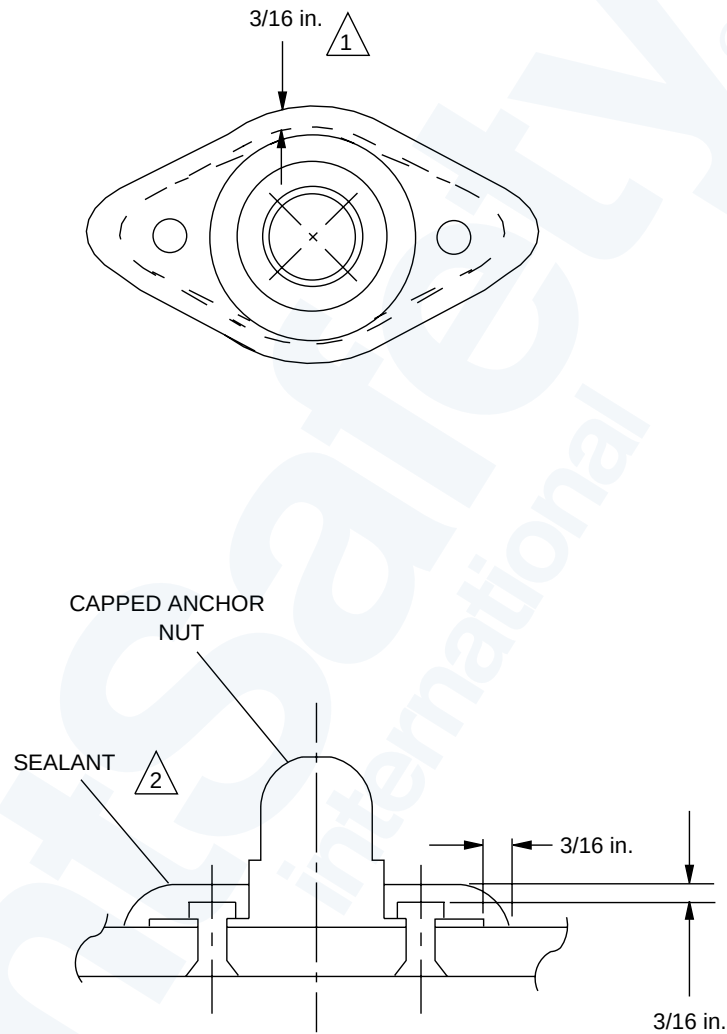
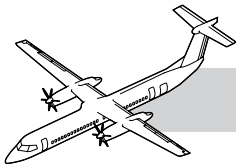


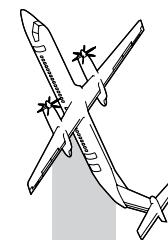
Figure 51-7. Wet Installation of Permanent Fasteners



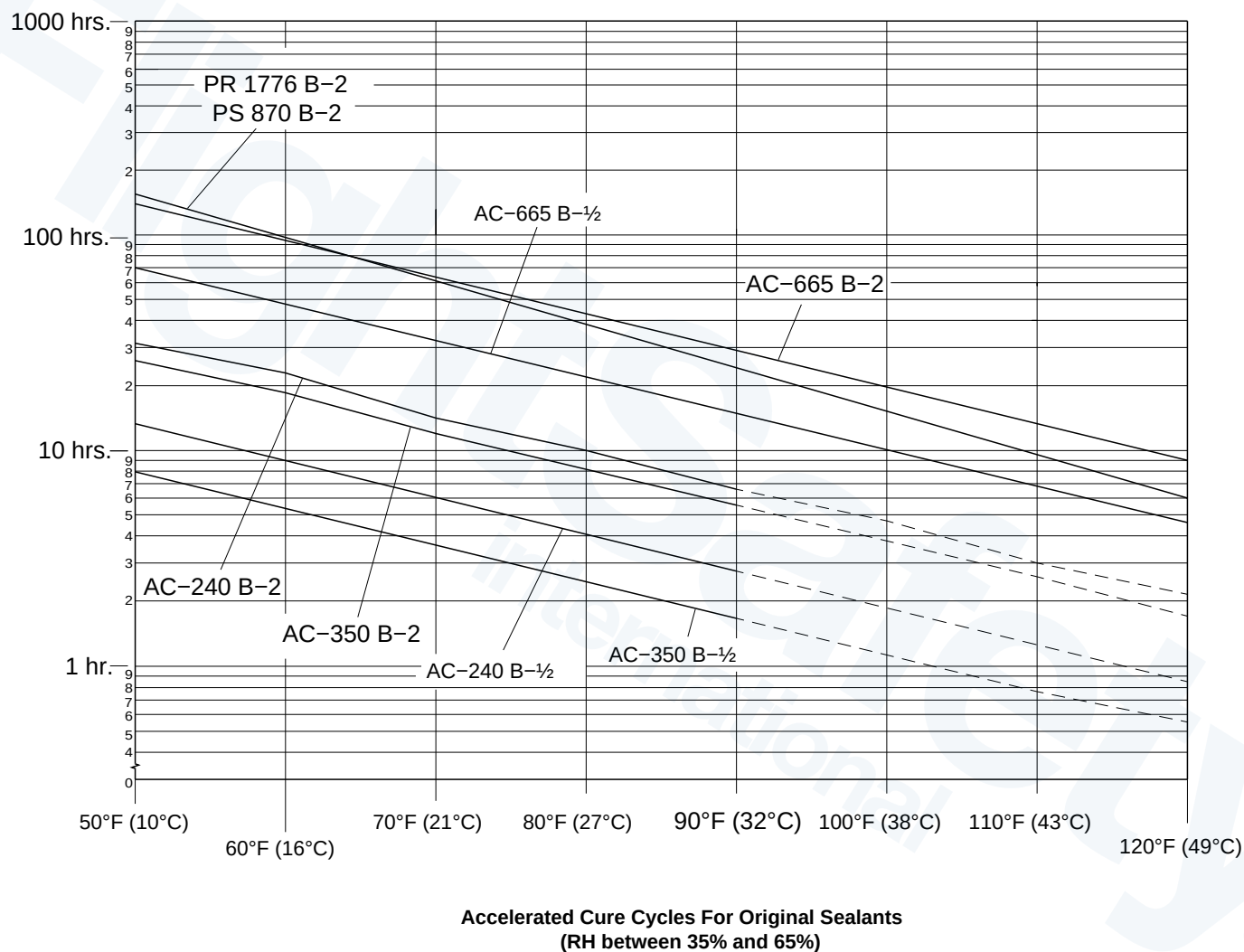
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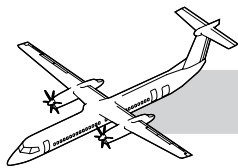
- 1 Sealant should extend 3/16 in. outward from the plate edge.
- 2 The sealant around the fuel tank access panel anchor nuts (Sta. Yw 42.000 to Yw 191.200) is classified as Critical Design Configuration Control Limitation (CDCCL) items with the purpose of providing protection in the event of the lightning strike. For the details refer to PSM 1-84-7, Part 2.

Figure 51-8. Dome Sealing Anchor Nut

**NOTE**

This figure is a general guideline on the curing behavior of sealants with respect to temperature variations. Wide variations in relative humidity (RH) will also affect curing times. This guideline should only be applied in the range of 35% to 65% RH. Broken lines identify sealants that require an adhesion promoter when cure is forced over 90°F (32°C).

**Figure 51-9. Accelerated Cure Cycles for Original Sealing Compounds**



Alternate Fast Cure Sealant

Refer to Figure 51-9. Accelerated Cure Cycles for Original Sealing Compounds.

Alternate sealants can be used in the fuel cells when faster curing times are required to reduce the turn-around time on A.O.G aircraft. Time to fuel is based on sealant applied at 16 to 24°C. For every 12°C decrease in temperature, the time to fuel will double. Example: At 10°C a 6 hrs time to fuel is now 12 hrs. Above 29°C the application time of these materials is down to less than 15 minutes. The alternate sealant can be used where the original sealant is identified in the documentation.

NOTE

All De Havilland acceptable substitutes are based on Sealant Manufacturer's data or appropriate military specifications, See SRM for proper application.

CAUTION

The alternate sealants mentioned are not permitted for use in high temperature applications.

NOTE

When using some sealants (eg. Class B sealant) it is strongly recommended that the aircraft substrates be primed with "Adhesion Promoter".

CAUTION

Take care to make sure that sealant is applied within the time specified by the manufacturer of the sealant due to rapid cure time of some sealants. This is highly important when using "1/4" or "1/2" type of sealant.

NOTE

This figure is a general guideline on the curing behavior of sealants with respect to temperature variations. Wide variations in relative humidity (RH) will also affect curing times.

This guideline should only be applied in the range of 35% to 65% RH.

Broken lines identify sealants that require an adhesion promoter when cure is forced over 90°F (32°C).

Dispatch of the Aircraft with Uncured or Missing Sealant

To allow aircraft to return to service prior to the completion of the curing cycle of Aerodynamic/Weather sealant applied to various surfaces.

Discussion

The application of High Speed Tape over the uncured sealant is acceptable, subject to the following:

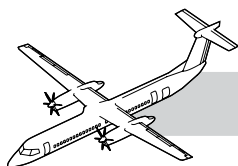
Limitations

Tape may not be applied to the wing and horizontal stabilizer leading edges: If it loses adhesion during operation it could disturb the airflow, possibly changing the aircraft stall speeds.

Tape may not be used around engine air inlets, static ports, or NACA vents.

NOTE

It is permitted to operate the aircraft with the leading edge sealant removed from up to one wing leading edge, and one vertical stabilizer leading edge, and one horizontal stabilizer leading edge simultaneously from an aircraft.

**NOTE**

The above surfaces may be left unsealed for a maximum of 50 flight hours. At or prior to reaching the 50 hour limit, the leading edge(s) are to be removed; the areas must be inspected, cleaned, sealed and the sealant permitted to fully cure.

Recommendations

To improve the chances of the sealant fully curing:

- Seal with an approved sealant with the shortest cure time identified in the documents
- Allow the sealant to reach a “Tack Free” condition on the outer surface before tape application
- Ensure the area of adhesion for the tape is clean and dry
- Ensure the tape has properly adhered to the structure and component surfaces.
- Cover the exposed sealant with a suitable tape
- Keep the number of seams to a minimum and consider the airflow direction when applying the tape
- Tape is to be inspected daily for proper adhesion until removal
- If precipitation is encountered during flight, the adhesion should be confirmed prior to next flight
- Areas exhibiting signs of adhesion loss must be repaired prior to flight
- Tape is to be removed at first opportunity upon reaching final sealant cure
- The sealed area is to be checked to confirm sealant has remained in place and fully cured no later than the first ‘L’ check after the tape has been removed and flight operations resumed.

NOTE

Tape which comes in contact with ice crystals, snow, etc. may develop P-static build up resulting in noise on the radios and/or in the headsets.

NOTE

Due to temperature, pressure and humidity changes the sealant will encounter during aircraft operation, it is recommended the full cure time be extended by 50%.

Tapes

Aluminum Foil Tape: 3M 425 or 427, Permacell 112, Mystic 7453 or Matrix Technology A662.

Aluminum Foil Tape with Conductive Adhesive: 3M 1170.

NOTE

P-static build up is less likely to develop using the Aluminum Foil conductive adhesive tape.

CAUTION

Aluminum Foil Tape is not suitable for around radiating devices such as V/Stab leading edges with HF antenna installation, radome or any antenna.

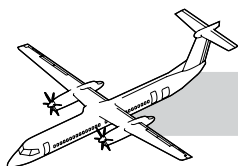
3M 8403, Matrix Technology M851, Airtech International Flashbreaker 1 or 2, or Richmond Aircraft Products Flashtape 1.

NOTE

Polyester/Polyvinyl Tape is suitable for all locations.

CAUTION

Polyester/Polyvinyl Tape is a possible source of P-static.



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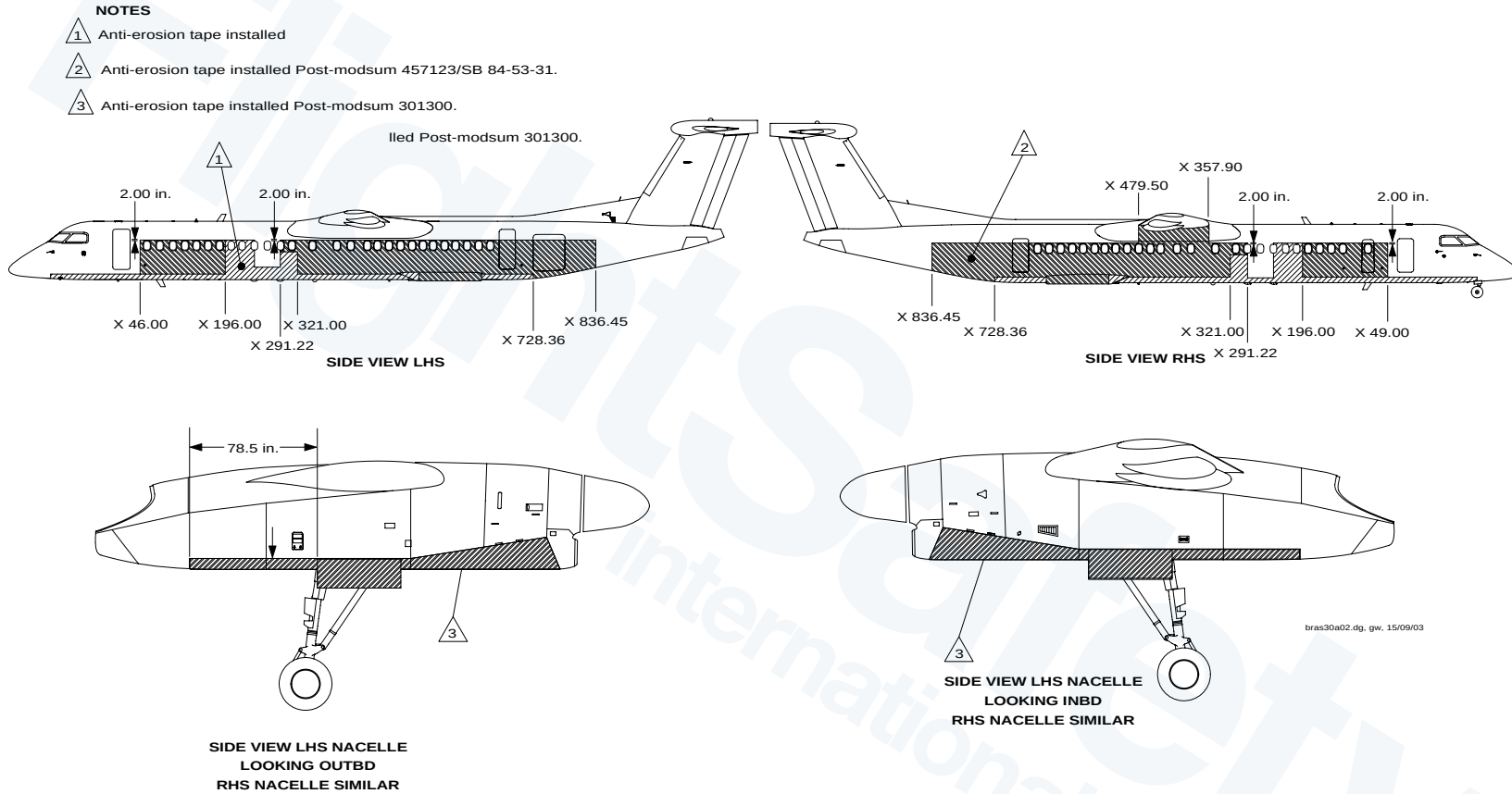
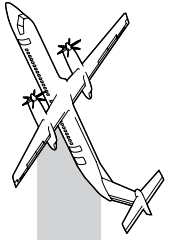
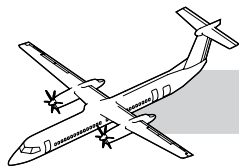


Figure 51-10. Unpaved Runway Protective Tape Application



51-25-00 APPLICATION OF AIRCRAFT PAINT

INTRODUCTION

The mating surfaces between the fuel tank access panel and the wing skin (Sta. YW 191.2 to YW 425.019) is designed with metal-to-metal interface electrical bonding that must not be primed or painted. Only chemical conversion coating is permitted and no other finishes are permitted. The metal-to-metal interface is a fuel tank safety critical item and is classified as a Critical Design Configuration Control Limitation (CDCCL) item. Its integrity must be maintained to make sure that unsafe conditions do not develop by maintenance, modification or repair. For the details, refer to PSM 1-84-7, Part 2.

GENERAL

Refer to Figure 51-10. Unpaved Runway Protective Tape Application.

On aircraft incorporated with post modsum 301300 or/and post modsum 457123/SB 84-53-31, the protective coating consists of polyurethane tape (as an alternative to coating of the fuselage with an anti-erosion paint) followed by the paint scheme colors.

Removal of Damaged Runway Protective Tape

Protective tape applied to the aircraft structure must be free from any damage, wrinkles, folds and air bubbles. If there is any damage to the protective tape remove that section of protective tape from the aircraft structure. Recommended removal method for metallic surfaces: chemical stripper. Recommended removal method for composite surfaces: 3M rotary wheel.

CAUTION

Do not use metal scrapers or other sharp tools to remove. Metal scrapers or sharp tools can cause structural damage to the aircraft surface.

Use a plastic scrapper to remove protective tape if required. Clean the surface with solvent. Re-apply the protective tape to the area.

Where the optional unpaved runway protective coating (F36) is specified with a polyurethane external paint system, such coatings are applied under the polyurethane system as designated on the exterior paint scheme drawing.

When unpaved runway protective coating (F36) has been ordered for the aircraft to be painted with epoxy (F22) or acrylic lacquer (F26 or F27) external paint systems, consult the de Havilland materials technology for painting procedures.

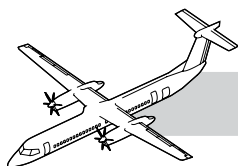
Where non-skid coatings are specified, clear F24 enamel is applied over the cured external paint. Non-skid material (walnut shells) is sprinkled over the first layer and another layer of F24 clear paint sprayed over the non-skid material.

All surfaces which have been primed are painted as follows:

- A tack or thin layer is applied
- It is air dried
- A full final layer is applied
- Let the paint cure.

Special Tooling

- Film thickness gauge
- Electrometer
- Micrometer or equivalent
- Spray gun.



Safety Precautions

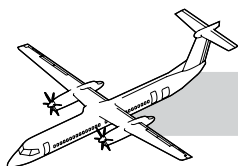
NOTES

CDCCL Requirement

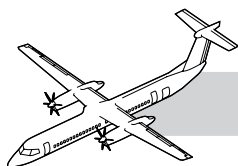
Do not paint the mating surface of the access panel and wing skin (STA. YW 191.2 TO YW 425.019). Only chemical conversion coating is permitted and no other finishes are permitted. The metal-to-metal interface between the access panel and wing skin is necessary to provide electrical bonding. Painting of the mating surface will eliminate the electrical bonding. This can possibly cause an ignition source inside the fuel tank in the event of a lightning strike.

Do not damage the layer of primer or the non-metallic surface.

Make sure that you do not penetrate through the primer coating to avoid damaging the pretreatment underneath.



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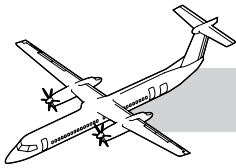
**Non-conductive finishes that you can remove:**

NON CONDUCTIVE FINISH	BASE METAL	COLOR
Primers	ALL	Yellow or green
Topcoats	ALL	As specified
Anodic coating	Aluminium	Grey/yellow
Phosphate	Steel	Dull grey
Stains and dyes	All	Full range of colors
Fuel tank coating	All	Yellow or blue green
Adhesive on wire mesh	Aluminium wire	Not transparent

Non-conductive finishes that you cannot remove:

NON CONDUCTIVE FINISH	BASE METAL	COLOR
Alodine 600, 1200	Aluminum	Iridescent gold or brown
Alodine 1000	Aluminum	Without color
Clear Iridite	Aluminum	Without color
Cadmium plate	Steel	Bright iridescent bronze
Chrome plate	Steel	Bright metallic
Nickel plate	Steel	Satin chrome bright metallic
Tin plate	Copper or steel	Bright metallic

**Figure 51-11. Bonding Strap**



51-80-00 ELECTRICAL BONDING

INTRODUCTION

Electrical Bonding ensures electrical conductivity between the components on the aircraft. The bonding resistance check has to provide a low resistance path between components by means of straps, or surface contact. The bonding resistance measurement check is done after a component (LRU) is changed.

The general precautions to electrically bond components are as follows:

- All paint, primer, stains, and other finishes are removed from the area that is to be electrically bonded
- The component that needs to be electrically bonded, is bonded directly to the structure and not through other bonded parts
- Electrically bonded components which are put in series are permitted for plumbing lines
- When more than two components are bonded in series, the ends of the components are bonded to the structure
- The electrical bonded area is approximately 1/16 in. larger than the diameter of the washer that touches the area
- The metal surfaces must touch each other
- If the procedures give instructions to apply a layer of Corrosion Preventative Compound (CPC) to the shank of a conductive bolt, make sure that the CPC does not get under the head of the bolt.

Bonding Check - Resistance

Refer to Figure 51-11. Bonding Strap.

CAUTION

Make sure the bonding resistance tester has a valid calibration decal.

WARNING

DO NOT USE A BIDDLE TESTER IN AREAS WHERE FUEL VAPORS ARE PRESENT, USE A LOW POWER DIGITAL MULTIMETER TO AVOID ANY POTENTIAL FOR SPARKING AT THE TEST POINTS. MAKE SURE THE AIRCRAFT IS ELECTRICALLY GROUNDED.

Procedure

Use low resistance test equipment with a resistance range of 0 - 10 milliohms. The resistance value must be less than specified. If the bonding check resistance value exceeds specification:

- Remove the component and check that all bonding point surfaces are thoroughly cleaned before the installation of the component and bonding jumper
- Make sure component and all bond connections are tightened to the torque value specified in the applicable component installation procedure
- Inspect applicable bonding cable for damage or insecure wire braiding
- Replace bonding cable if damage is found.

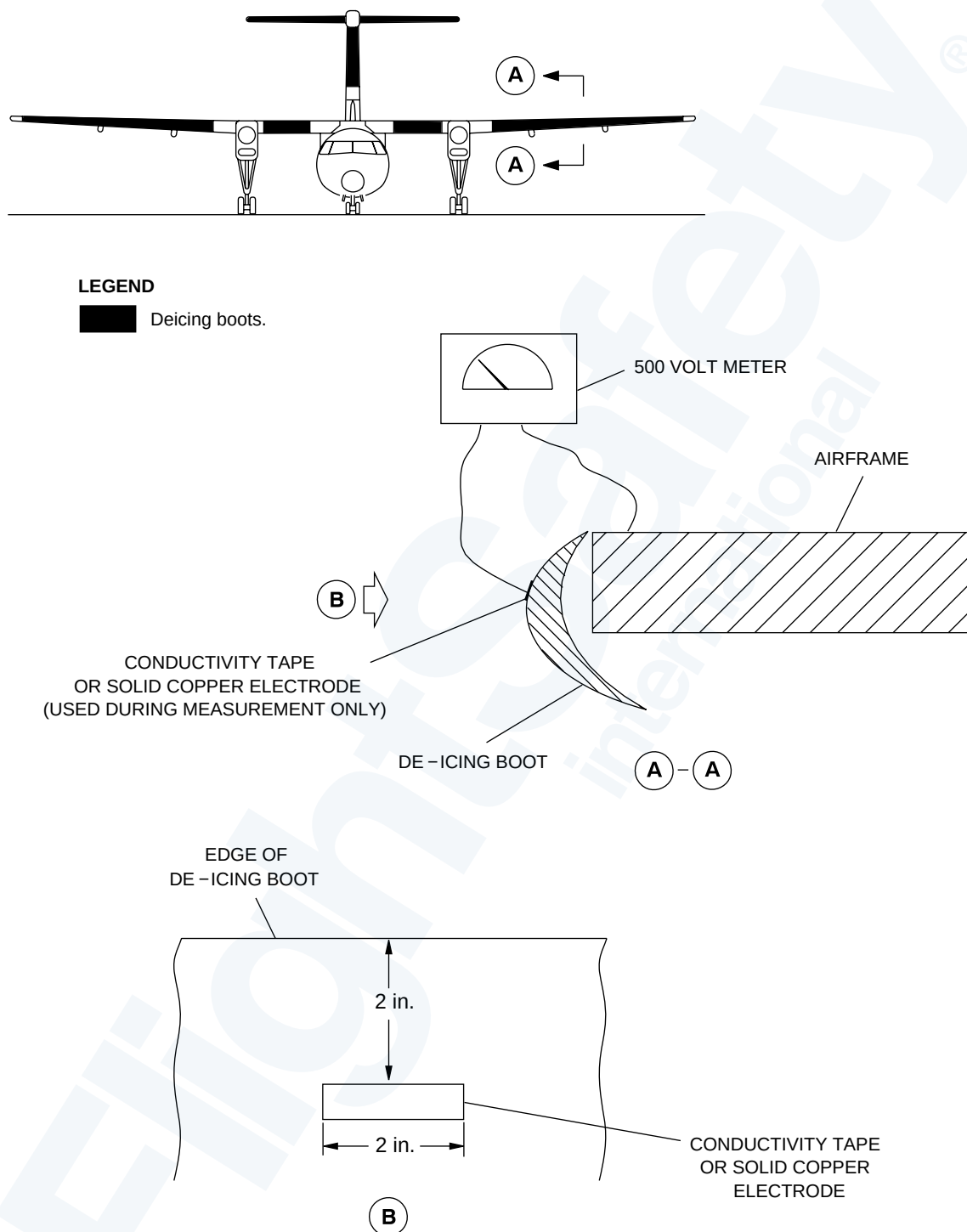
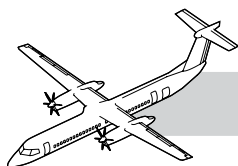
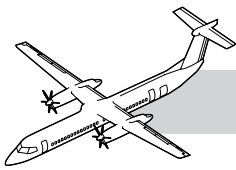


Figure 51-12. Resistance Measurement - De-Icing Boot/Airframe



Resistance Measurement

Refer to Figure 51-12. Resistance Measurement - De-Icing Boot/Airframe.

General

Resistance measurements on aircraft structural components and aircraft components.

- For the Line Replaceable Units (LRUs) with an electrical connector, measure the resistance from the electrical connector to the electrically conductive surface on the aircraft structure
- For the LRUs without an electrical connector, measure the resistance from the unit to the electrically conductive surface on the aircraft structure
- For the hydraulic tubes, measure the resistance from the hydraulic tubes to the electrically conductive surface on the aircraft structure
- For the hydraulic equipment with no electrical connection, measure the resistance from the electrical conductive surface on the housing to the electrically conductive surface on the aircraft structure

CAUTION

All the electrical equipment must be tested with the mating electrical connectors removed. The connectors must be installed after the test is completed.

Surface Preparation for Electrical Bonding:

- SRM 51-80-00-02-01 Grounding and Bonding of Metallic Components
- SRM 51-80-00-02-02 Grounding and Bonding of Non-Metallic.

Use a one megaohm multimeter to measure the electrical resistance of the conductive layer.

- Put the multimeter probes tightly against the conductive layer, approximately 12.00 in. (304.80 mm) apart

NOTE

If the surface is less than 12.00 in. (304.80 mm) wide, then put the multimeter probes as far apart as possible.

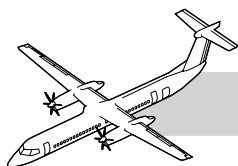
- The maximum permitted resistance value for the conductive layer is 300,000 ohms
- Take a sufficient number of measurements to get a representative sample of the electrical resistance of the conductive layer

NOTE

You must reject the component if the resistance measurement is larger than the maximum permitted value.

Resistance measurement between the de-icer boots and the airframe:

- Use a 500 VDC megger tester to measure the resistance between the de-icer boot and the airframe
- Use a conductive tape approximately 0.5 in. by 2.0 in. (12.7 mm by 50.8 mm) to measure the electrical resistance of the de-icing boot.

**NOTE**

Solid copper electrode could be used as an alternative to the conductive tape.

- Put the conductive tape parallel to the edge of the de-icing boot, approximately 2.0 in. (50.8 mm) from the edge of the boot
- The measured resistance should be less than one megaohm.

NOTE

You must reject the component if the resistance measurement is larger than the maximum permitted value.

Application of HumiSeal[®] 1B15

Cleanliness of the substrate is of extreme importance for the successful application of the coating.

Surfaces must be free of moisture, dirt, wax, grease, flux residues and all other contaminants.

Contamination under the coating could cause problems that may lead to assembly failures.

- Make sure that all the electrical bonding hardware are fully tightened.
- Make sure you solvent clean an area approximately 50% larger than the sealant contact area.
- Extend the sealant approximately 0.125 inches beyond the area stripped of protective coating.
- Make sure that the sealant does not come between hardware where metal to metal contact is necessary.
- Before the sealant cures, use the plastic scraper and lint-free cloth moistened with solvent to remove all unwanted sealant.

Relief for Continued Operation with Broken/Missing Lanyards for a Period of 600 Flight Hours

Continued operation with missing/broken lanyards for 600 flight hours.

- Nose O₂ Servicing Port - Puritan Bennett Cap and Chain
- Hydraulic HPU connection and Service Ports
- Air Conditioning ground connection door
- Lanyard Assembly
- Empennage Access Door
- Refuel/Defuel cap and Access Door.

Temporary Repair for Continued Operation with a Broken Bonding Jumper

This is applicable to the bonding jumpers on the:

- Main landing gear assembly
- Nose landing gear assembly
- Main landing gear doors
- Nose landing gear doors.

Procedure

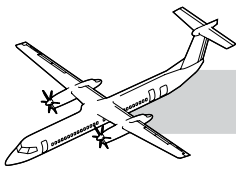
Cut the remaining wire strands with a wire cutter, from each end of the bonding jumper, flush to the end of the terminal

Inspect to make sure that the terminal ends do not touch on the adjacent structure or equipment.

NOTE

This repair is valid for 25 flight hours.

At or before 25 flight hours have elapsed, replace the affected bonding strap with a new part in accordance with AMM.



Bonding Check of the Ice Detector Probe (IDP)

The bonding resistance measurement check is done after an IDP is changed.

Tools and Equipment

- Biddle Instruments digital low resistance ohmmeter
- DLRO milliohm meter.

NOTE

Make sure the bonding resistance tester has a valid calibration decal.

Procedure

Disconnect the connector of the IDP. Connect a long test lead to the seat rail through the main cabin door to the test meter.

NOTE

If a long test lead is not available connect the test lead to a ground stud near the IDP.

Connect the second test lead from the test meter to the receptacle back shell of the IDP. Set the test meter on the lowest scale setting, to ensure maximum applied test current. Measure and record the resistance between the receptacle back shell and the seat rail or ground stud near the IDP to be tested.

NOTE

The resistance value must be less than or equal to 10 milliohms.

If the bonding check resistance value is less than or equal to the value specified then connect the IDP.

If the resistance value is more than the value specified:

- Rework the IDP flange mounting area
- Make sure all bond check connections are tight
- Repeat the bonding check.

Reject the IDP if the resistance measurement is more than the value specified.

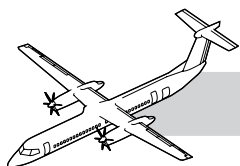
GENERAL

The general precautions to bond electrical jumpers are as follows:

- Make sure the bonding jumpers are not longer than 3.00 in. (76.20 mm)
- Do not compress the electrical jumper connections when you install them through material which is not metal
- When you install the bonding jumpers, make sure that parts that move have sufficient distance to move
- When you install the bonding jumpers, make sure that vibration during usual operation does not break or loosen the connection.

The general precautions for electrically bonded clamps are as follows:

- When you install the bonding clamps, make sure that the conduit or hose is not damaged
- You must only use metal clamps
- When you install the bonding clamps, make sure that vibration during usual operation does not break or loosen the connection.



The general precautions for plumbing lines are as follows:

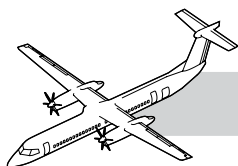
NOTES

- If plumbing lines transmit fluids and have a resistance of more than 1 ohm, you must electrically bond them
- You must electrically bond AN fittings that have the anodic layer removed
- If the distance between plumbing lines and the aircraft structure is less than 144.00 in (3.66 m) you must install bonding jumpers
- You must know that the tube connectors that follow are a conductive part of the plumbing line:
 - Wiggins W700DE
 - Wiggins W991 A (DE)
 - Hydraflow equivalent.

Damage Reporting

NOTE

For all structural damage reporting, there is a damage reporting form supplied by Bombardier.



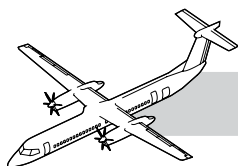
51-00-00 SPECIAL TOOL & TEST EQUIPMENT

- MIL's V 16 Vacuum Pump
- None specified Viscometer
- GSB2400001 Digital Multimeter - Hand Held
- GSB2400002 Milliohm Meter
- GSB2400012 Insulation Resistance Tester
- GSB5180001 or GSB5180005 Bonding Tester
- HP 4328A milliohmmeter
- Biddle Instruments digital low resistance ohmmeter, DLRO #247001
- Biddle Instruments digital low resistance ohmmeter, DLRO #247000 or equivalent milliohm meter
- GSB2400002 Milliohm meter, BCD M1 Milliohm meter or equivalent
- GSB5180001 Tester - Electrical Bonding, 120/230 VAC, Megger 247001, Megger DLRO 10 or equivalent bonding test meter
- SB2400002 Milliohm meter, BCD M1 Milliohm meter or equivalent

51-00-00 MAINTENANCE PRACTICES

Refer to the Bombardier AMM PSM 1-84-2 for details on these maintenance procedures:

- AMM 51-21-01-340-801: Application of General Electric PSA-529/SRC - 18 Two Part Adhesive.
- AMM 51-21-06-340-801: Application of RTV 730/FRV1106 Fluorosilicone Adhesive.
- AMM 51-21-11-340-801: Application of EC 1300L Neoprene Base Adhesive.
- AMM 51-21-16-340-801: Application of Bostik 7132/Boscodur No.22 Adhesive.
- AMM 51-21-16-340-802: Application of Bostik 1142 Adhesive.
- AMM 51-21-21-340-801: Application of Aquabind Water Based Adhesive.
- AMM 51-21-26-340-801: Application of RTV Silicone Adhesive.
- AMM 51-21-31-340-801: Application of 3M EC2216 Two Part Epoxy Adhesive.
- AMM 51-21-36-340-801: Application of Epoxy Adhesive.
- AMM 51-21-41-340-801: Application of 3M EC1357N Neoprene Base Adhesive.
- AMM 51-21-46-340-801: Application of Loctite Superbond 430 Adhesive.
- AMM 51-21-51-340-801: Application of Epoxy, Fast-Weld #10.
- AMM 51-23-00-390-806: Guidelines for the Dispatch of Aircraft with Uncured Sealant.
- AMM 51-23-00-390-807: Temporary Allowance for Continued Operation with Missing Sealant on the External Access Panels.
- AMM 51-23-01-390-801: Mixing and Handling of Two-Part Sealants.



- AMM 51-23-01-390-802: Application of AC-240 Class B Sealant.
- AMM 51-23-11-390-801: Application of Loctite Sealant.
- AMM 51-23-16-390-801: Application of Integral Fuel Tank Sealant.
- AMM 51-23-16-390-802: Application of Sealant on the Fuel-Tank Vent Line Support Bracket and Attachment Hardware, Sta.YW 42.00.
- AMM 51-23-18-390-801: Application of Sealant to Removable Parts.
- AMM 51-23-21-390-801: Application of Sealant on the External Paint Edges.
- AMM 51-23-23-390-801: Application of Devcon Putty.
- AMM 51-23-26-390-801: Application of Sealant for High Temperature Zones and Firewalls.
- AMM 51-25-01-390-801: Application of F13 Corrosion Preventive Compound.
- AMM 51-25-06-390-801: Application of F35 Fuel Vapor Barrier Epoxy Coating.
- AMM 51-25-11-390-801: Application of DSC216 Fluorocarbon Dispersion Coating.
- AMM 51-25-16-390-801: Manual Application of C1 Chemical Coating.
- AMM 51-25-21-640-801: Application of Molykote 321R Dry Film Lubricant.
- AMM 51-25-26-370-801: Application of F19 and F45 Epoxy Polyamide Primer.
- AMM 51-25-31-370-801: Application of Aircraft Paint.
- AMM 51-25-36-370-801: Application of F31 and F34 Carbon Filled Polyurethane Enamel.
- AMM 51-25-41-370-801: Application of F23 Epoxy-Amine Intermediate Primer.
- AMM 51-25-46-370-801: Application of F41 Anti-Static Coating.
- AMM 51-25-51-340-801: Application of F33 Surface Finishing Compounds.
- AMM 51-25-56-390-801: Application of Release Agent #13.
- AMM 51-25-61-390-801: Application of Polyurethane Protective Tape (PPT) and Polyurethane Protective Radome-Boot (PPRB).
- AMM 51-25-61-390-802: Application of Polyurethane Protective Tape (PPT) for Unpaved Runway Protection.
- AMM 51-27-01-110-801: Solvent Cleaning.
- AMM 51-27-02-110-801: Cleaning of the Seat Rail.
- AMM 51-27-06-110-801: Removal of Grease with Vapors.
- AMM 51-80-00-910-801: Electrical Bonding Safety Precautions.
- AMM 51-80-00-120-801: Surface Preparation for Electrical Bonding.
- AMM 51-80-00-400-801: Installation of Electrical Bonding Devices.
- AMM 51-80-00-760-801: Resistance Measurement.
- AMM 51-80-00-760-802: Bonding Check - Resistance.
- AMM 51-80-00-390-801: Application of Electrical Bonding Sealant.
- AMM 51-80-00-760-805: Bonding Check of the Ice Detector Probe (IDP).